MAT067 University of California, Davis Winter 2010

Midterm Exam

(Wednesday, February 10)

Problem 1 (15 points)

a) Suppose we are given a complex number $z = x+iy$ where $x, y \in \mathbb{R}$. Write the definitions of the complex conjugate $\overline{z}$ and the modulus $|z|$.

b) Using only part (a), prove that for nonzero z the number $w = \dfrac{\overline{z}}{|z|^2}$ is an inverse of z.

c) Prove that $e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$ using only the polar form definition of $e^{i\theta}$.

d) Find all solutions over $\mathbf{C}$ to $z^3 = -1$.

Problem 2 (15 points)

Let V be a vector space over $\mathbf{F}$.

a) State the three conditions for a subset $U \subseteq V$ to be a subspace of V.

b) Let U_1 and U_2 be subspaces of V. Which of the above conditions prevents $U_1 \cup U_2 = \{v \in V \mid v \in U_1 \text{ or } v \in U_2\}$ from being a subspace of V?

c) Let U_1 and U_2 be subspaces of V. Show that if $V = U_1 \oplus U_2$ (direct sum) then $U_1 \cap U_2 = \{0\}$.

d) Let $V = \mathbf{C}^4$ and $U_1 = \{(a, b, c, d) \in V : d = 0\}$. Show that U_1 is a subspace of V.

e) Show that V is not the direct sum of U_1 and U_2 where $U_2 = \{(a, b, c, d) \in V : a + 2b + 3c + 4d = 0\}$.

Problem 3 (15 points)

Suppose V and W are vector spaces over $\mathbf{F}$ and $T : V \to W$ is linear.

a) Write the definitions of $\text{null}(T)$ and $\text{range}(T)$.

b) Suppose we have found a subspace U such that $V = \text{null}(T) \oplus U$. Show that $\text{range}(T) = \{T(u) : u \in U\}$.

c) In the setting of part (b), show that $\dim(\text{range}(T)) = \dim U$.

d) Using the formula from part (b) and/or (c), or other methods, find $\dim(\text{null}(T))$ for the map $T : \mathbf{R}^4 \to \mathbf{R}^3$ given by

$$T : \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \longmapsto \begin{pmatrix} 1 & 1 & 4 & 3 \\ 2 & 1 & 3 & 2 \\ 3 & 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}.$$